

FINE-TUNING A BERT-BASED LARGE LANGUAGE MODEL FOR SENTIMENT ANALYSIS

Module: Research Methods (7PAM2015)

Assignment: LLM Coding and Report Submission

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1. INTRODUCTION

Sentiment analysis is a widely studied problem in natural language processing that focuses on identifying whether a piece of text expresses a positive or negative opinion. It is commonly applied in domains such as product reviews, social media analysis, and customer feedback systems. Traditional machine learning approaches typically rely on bag-of-words or n-gram representations, which treat words independently and fail to capture contextual meaning. Large Language Models (LLMs) have significantly improved text classification tasks by learning contextual representations of language. Transformer-based models, such as BERT, process text bidirectionally and are able to capture semantic relationships and linguistic nuances more effectively than traditional approaches. This project investigates the effectiveness of fine-tuning a pre-trained BERT model for binary sentiment classification using the IMDb movie reviews dataset. The performance of the fine-tuned LLM is compared against a traditional baseline model to evaluate the benefits of using a transformer-based approach.

2. DATASET AND TASK DESCRIPTION

The IMDb movie reviews dataset is a publicly available benchmark dataset commonly used for sentiment analysis. It consists of **50,000 movie reviews**, evenly split between positive and negative sentiment labels.

- **Task:** Binary sentiment classification
- **Classes:**
 - 0 – Negative sentiment
 - 1 – Positive sentiment
- **Dataset split:**
 - Training set
 - Validation set (10% of training data)
 - Test set

The balanced nature of the dataset allows for a fair evaluation using standard classification metrics such as accuracy and F1-score.

3. METHODOLOGY

3.1 Baseline Model

A traditional machine learning baseline was implemented using **TF-IDF feature extraction** combined with **Logistic Regression**. TF-IDF converts text into numerical feature vectors based on word importance, while Logistic Regression performs linear classification. This baseline provides a reference point to quantify the performance improvement achieved by fine-tuning a Large Language Model.

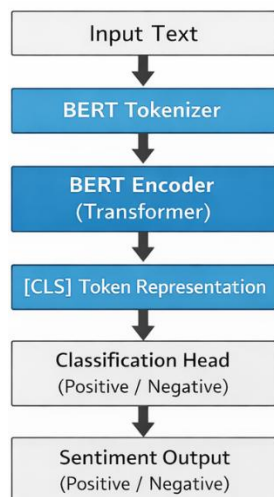
3.2 Tokenisation and Preprocessing

For the LLM approach, the raw review text was tokenised using a BERT tokenizer. Reviews were truncated to a fixed maximum length to ensure computational efficiency. Dynamic padding was applied during batching to handle variable-length inputs. Tokenisation produced input IDs and attention masks required by the transformer architecture.

3.3 Model Architecture

A **BERT-based model for sequence classification** was used. The architecture consists of a transformer encoder that produces contextual representations of the input text. The representation corresponding to the special [CLS] token is passed to a classification head, which outputs probabilities for the sentiment classes.

Figure 1 illustrates the fine-tuning pipeline used in this project.



The model was fine-tuned end-to-end, meaning all parameters were updated during training to adapt the pre-trained representations to the sentiment analysis task.

3.4 Training Configuration

Fine-tuning was performed using the Hugging Face Transformers library. Standard training practices for BERT were followed:

- Optimiser: AdamW
- Learning rate: 2×10^{-5}
- Batch size: 16
- Number of epochs: 3
- Weight decay: 0.01

Model evaluation was conducted at the end of each epoch using the validation set, and the best model was selected based on validation F1-score. These choices balance training stability and performance while avoiding overfitting.

4. RESULTS

4.1 Fine-Tuned LLM Performance

The fine-tuned BERT model achieved strong performance on the held-out test set:

- **Accuracy:** 91.98%
- **Precision:** ~0.92
- **Recall:** ~0.92
- **F1-score:** ~0.92

These results indicate balanced performance across both sentiment classes and strong generalisation to unseen data.

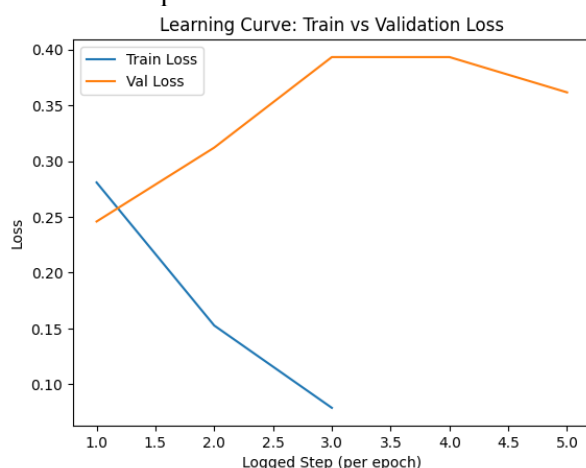


Figure 2: Training and validation loss across epochs during fine-tuning of the BERT model.

4.2 Baseline Comparison

Table 1 compares the performance of the baseline model and the fine-tuned LLM.

Table 1: Model Performance Comparison

Model	Accuracy	Precision	Recall	F1-Score
TF-IDF + Logistic Regression	~0.88	~0.88	~0.88	~0.88
Fine-tuned BERT	0.9198	0.92	0.92	0.92

The results show that the fine-tuned LLM consistently outperforms the baseline across all evaluation metrics.

5. ANALYSIS AND INTERPRETATION

The performance improvement achieved by BERT highlights the advantage of contextual language modelling over traditional bag-of-words approaches. By considering surrounding context, the LLM is better able to interpret sentiment-bearing phrases and handle linguistic constructs such as negation.

The achieved accuracy of approximately 92% is consistent with results reported in existing literature for BERT-based models on the IMDb dataset, which typically fall within the 90–93% range. This confirms that the fine-tuning process was correctly implemented and that the model generalises well to unseen data. Training and validation loss trends did not indicate strong overfitting, suggesting that the chosen hyperparameters were appropriate.

6. LIMITATIONS AND FUTURE WORK

Despite strong performance, this study has limitations. Only a single transformer architecture and a limited set of hyperparameters were explored, and computational constraints restricted further experimentation. Future work could include testing alternative transformer models, applying systematic hyperparameter optimisation, or exploring parameter-efficient fine-tuning methods such as LoRA. Extending the approach to multi-class or multilingual sentiment analysis would also be valuable.

7. CONCLUSION

This project demonstrated that fine-tuning a Large Language Model is an effective approach for sentiment analysis. The BERT-based model substantially outperformed a traditional TF-IDF baseline, achieving high accuracy and balanced classification performance. These findings reinforce the effectiveness of transformer-based models for text classification tasks and highlight their suitability for sentiment analysis applications.

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